

# Humzah Merchant

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## EDUCATION

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### Brown University

PhD in Computer Science – Advised by Profs. Randall Balestriero and Bradford Levy

Providence, RI

Expected May 2031

### University of Chicago

BA in Statistics – Advised by Prof. Bradford Levy

Chicago, IL

Expected June 2026

## CONFERENCE PAPERS

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### Divergence Decoding: Targeted Unlearning via Auxiliary Models – HM, BL

International Conference on Machine Learning 2026

- Motivated by earlier work on addressing look-ahead bias—which is a special case of unlearning—applied method of debiasing logits more broadly to unlearning benchmarks and found that it is state of the art across most of them

### Towards Financial World Modeling – HM, AG, SM, RB, BL

Submitted to Neural Information Processing Systems (NeurIPS) 2026

- Exploring the current state of SSL on financial markets, we find that while supervised learning beats SSL on many core tasks, representations from SSL models generalize better to auxiliary downstream tasks.

## WORKSHOP PAPERS

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### A Fast and Effective Solution to the Problem of Look-ahead Bias in LLMs – HM, BL

NeurIPS 2025 Workshop: Generative AI in Finance – **Best Poster Award**

- Drew on product of experts and importance sampling to develop a method for reducing and/or eliminating look-ahead bias in LLMs. Demonstrated its efficacy across a variety of finance-specific tasks

### Forecasting With LLMs: Improved Generalization Through Feature Steering – HM, BL

Submitted to ICML 2026 Workshop: Forecasting as a New Frontier of Intelligence

- Amplifying features associated with time-awareness in LLMs can reduce reliance on information from after a cutoff specified at inference-time

### Optimizing Pre-Training of Tabular Foundation Models by Shaping Geometry – HM, SV, RB

Submitted to ICML 2026 Workshop: Foundation Models for Structured Data

- TFM's motivate well-distributed intermediate embeddings; although supervised pretraining does not naturally produce them, explicitly shaping early-layer geometry improves classification performance.

## EXPERIENCE

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### Chicago Booth School of Business

Research Assistant for Prof. Bradford Levy

Chicago, IL

October 2024 – Present

### Apollo Global Management

Software Engineering Intern (*Received Return Offer*)

El Segundo, CA

May 2024 – August 2024

### USC Marshall School of Business

Research Assistant for Prof. Larry Harris

Los Angeles, CA

October 2023 – April 2024

### NASA Johnson Space Center - Summer Robotics Academy

Software Engineering Intern

Mechanical Engineering Intern

Houston, TX

June 2023 – August 2023

June 2022 – August 2022

## COMPETITIONS

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### UChicago Trading Competition

#### 1<sup>st</sup> Place, Case 2: Quantitative Research and Portfolio Optimization

- Around 160 contestants. Won the primary data and research challenge (Case 2) and built a news trading bot (Case 1)
- Writeup of winning solution: <https://nbviewer.org/github/HumzahM/UTC2025-Case2-Writeup/blob/main/case2.ipynb>